

The Atacama Desert

The world's highest desert and one of the driest places on Earth, a sterile, stony plateau full of salt flats, sand, and active volcanoes

Lying to the west of the Andes and parallel to the Pacific coastline, the Atacama forms a narrow strip of arid desert in northern Chile, averaging less than 100 miles (160 km) in width but extending about 620 miles (1,000 km) south from the border with Peru. With an average elevation of about 13,000 ft (4,000 m), this high plateau is, according to NASA studies, the driest place in the world. Some weather stations have never recorded a single drop of rain; even at lower levels, a dry spell of four years is common. The average rainfall in the desert's estimated 40,600 square miles (105,200 square km) range is just $\frac{1}{16}$ in (1 mm) a year.

In mountain shadows

The Atacama is a double rain-shadow desert, because two high mountain ranges, the Chilean Coast Range and the Andes, block incoming moisture from both Pacific and Atlantic Oceans. In this hyper-arid climate, nothing decays; Earth's oldest mummies, dating from around 7020 BCE, were found here, still beautifully preserved. However, on the eastern side, where the foothills of the Andes rise up, there is a small amount of rainfall in winter.

Although parts of the Atacama are considered "absolute desert," other areas are more hospitable. Fog zones called *lomas*, created at lower levels when clouds are trapped by mountains or coastal slopes, bring in enough moisture to sustain lichens, algae, and hardy cacti and shrubs. Birds, which are able to fly in and out as conditions dictate, are the most abundant wildlife: three species of flamingoes are found in the Atacama, feeding on algae growing in salt lakes, while hummingbirds and Peruvian song sparrows are seasonal visitors.

However, due to the high content of sulfur, copper, and mineral salts such as sodium nitrate in its soils, the desert is hostile to most animals apart from lizards, insects, and rodents—and the South American gray fox, which feeds on them. In fact, the Atacama's soil composition is so similar to samples taken from Mars that NASA uses it for testing space rovers.

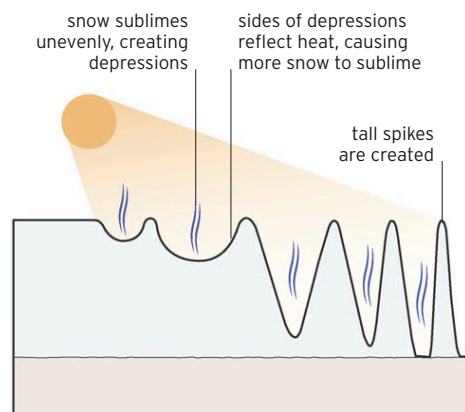


◀ HARDY MAMMAL

The highly adaptive South American gray fox, or *chilla*, copes with the Atacama's challenging conditions by hunting small animals in lower-lying fog zones.

HOW PENITENTES FORM

Penitentes are ice structures that usually form at altitudes of 13,000 ft (4,000 m) or more. Their formation is believed to be due to a combination of factors. When the Sun shines on snow, freezing temperatures prevent it from melting. Instead, it sublimates (changes directly from solid to gas without melting). Troughs form and reflect sunlight, raising the temperature enough to melt the surrounding snow. Strong winds freeze the melting snow into spikes.



Studies of sediments indicate that parts of the Atacama have had **no rain for more than 20 million years**